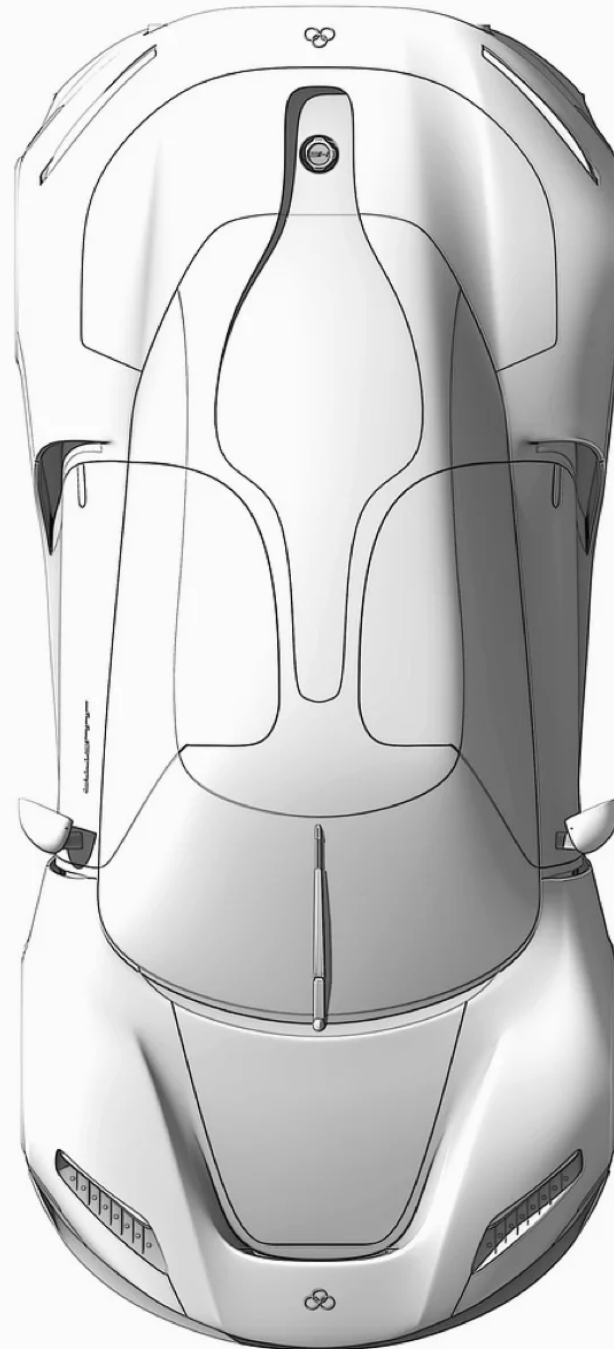


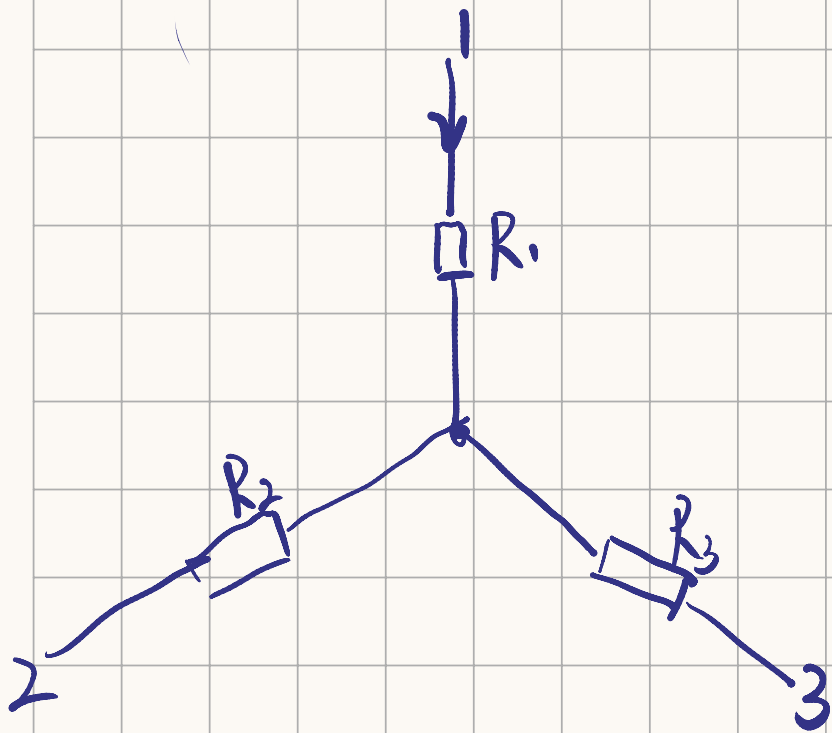
Title

Enjoy the most efficient handwriting experience! Taking notes on the go, whether for inspiration, ideas, knowledge learning, business insights, or even sketches...

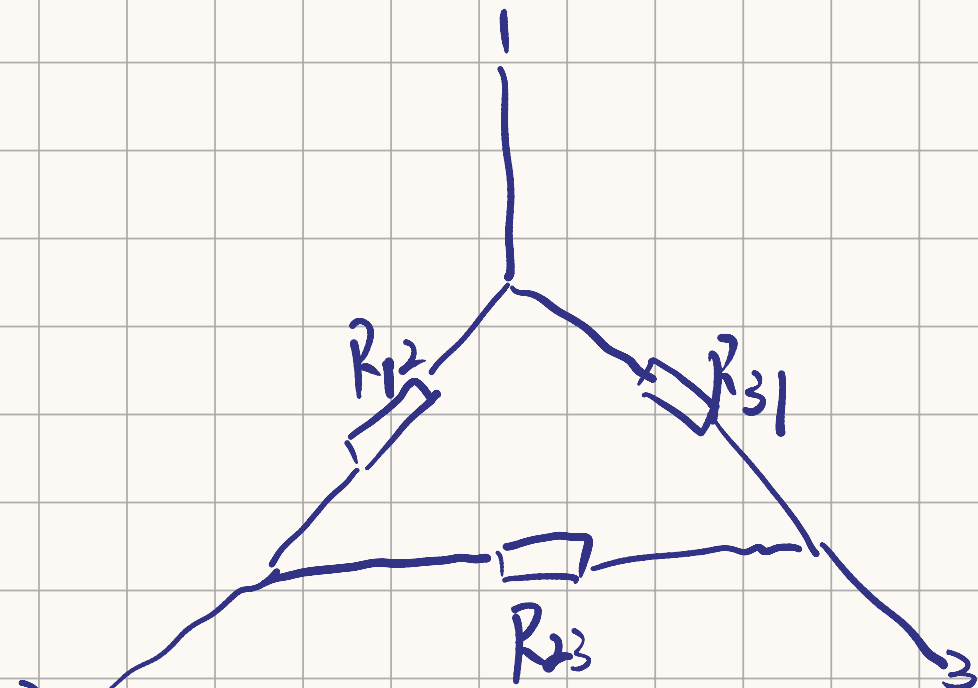


Creative notes

By reading we enrich the mind;
by writing we polish it.



$$\begin{cases} R_{12} = R_1 + R_2 \\ R_{23} = R_2 + R_3 \\ R_{31} = R_3 + R_1 \end{cases}$$



$$\begin{cases} R_{120} = R_{12} \parallel (R_{31} + R_{23}) = \frac{R_{12}(R_{31} + R_{23})}{R_{12} + R_{31} + R_{23}} \\ R_{230} = R_{23} \parallel (R_{12} + R_{31}) = \frac{R_{23}(R_{12} + R_{31})}{R_{12} + R_{31} + R_{23}} \\ R_{310} = R_{31} \parallel (R_{23} + R_{12}) = \frac{R_{31}(R_{23} + R_{12})}{R_{12} + R_{31} + R_{23}} \end{cases}$$

等效条件

$$R_{12Y} = R_{12L} = R_1 + R_2 = \frac{R_{12}(R_{31} + R_{23})}{R_{12} + R_{31} + R_{23}} = \frac{R_{12}R_{31} + R_{12}R_{23}}{R_{12} + R_{31} + R_{23}} \quad (1)$$

同理:

$$R_2 + R_3 = \frac{R_{23}(R_{31} + R_{12})}{R_{12} + R_{31} + R_{23}} = \frac{R_{23}R_{31} + R_{23}R_{12}}{R_{12} + R_{31} + R_{23}} \quad (2)$$

$$R_3 + R_1 = \frac{R_{31}(R_{12} + R_{23})}{R_{12} + R_{31} + R_{23}} = \frac{R_{31}R_{12} + R_{31}R_{23}}{R_{12} + R_{31} + R_{23}} \quad (3)$$

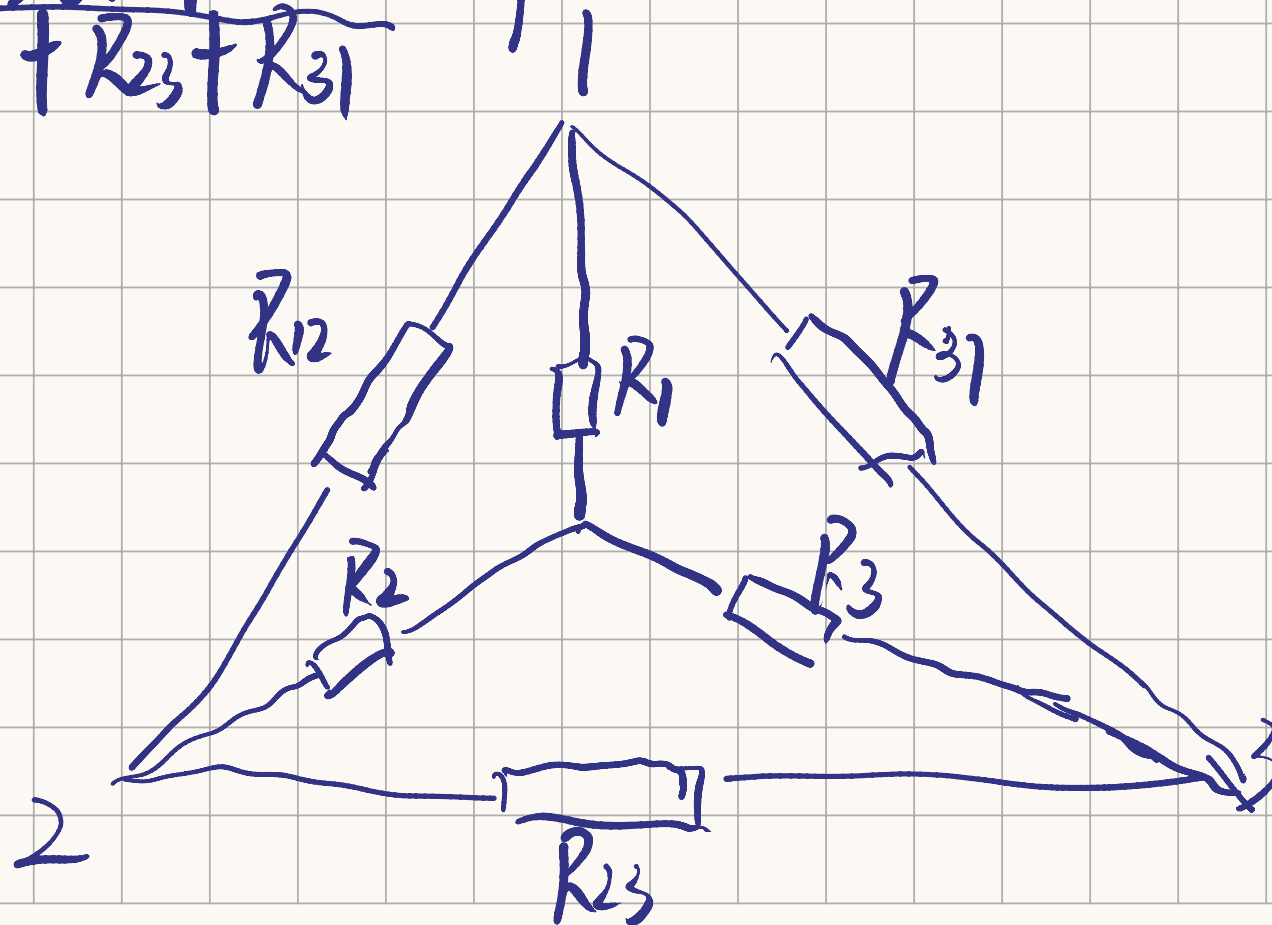
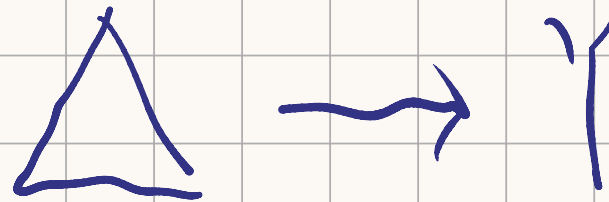
$$(1) + (3) - (2)$$

$$2R_1 = \frac{R_{12}R_{31} + R_{12}R_{23} + R_{12}R_{31} + R_{31}R_{23} - R_{23}R_{31} - R_{12}R_{23}}{R_{12} + R_{23} + R_{31}}$$

$$= \frac{2R_{12}R_{31}}{R_{12} + R_{23} + R_{31}}$$

$$\begin{cases} R_1 = \frac{R_{12} R_{31}}{R_{12} + R_{23} + R_{31}} \\ R_2 = \frac{R_{12} R_{23}}{R_{12} + R_{23} + R_{31}} \\ R_3 = \frac{R_{23} R_{31}}{R_{12} + R_{23} + R_{31}} \end{cases}$$

同理

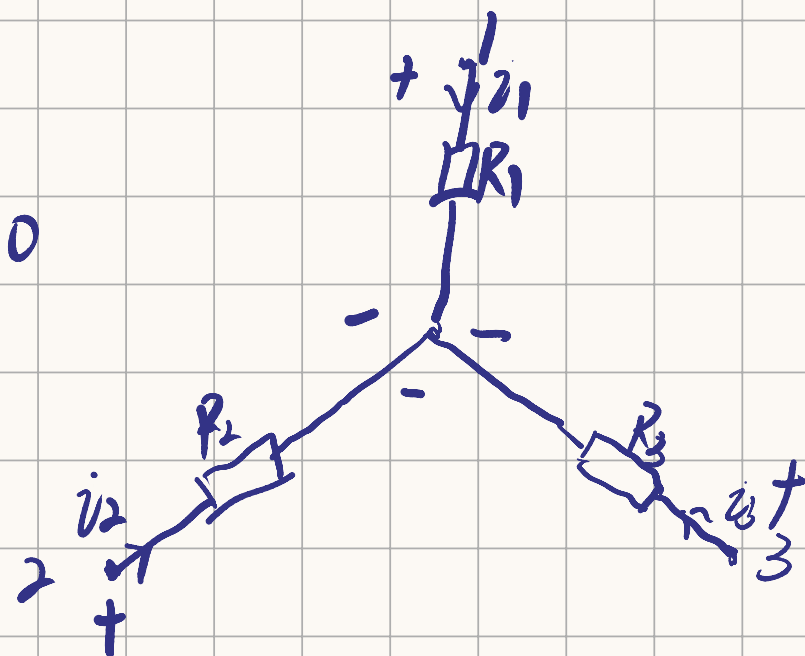
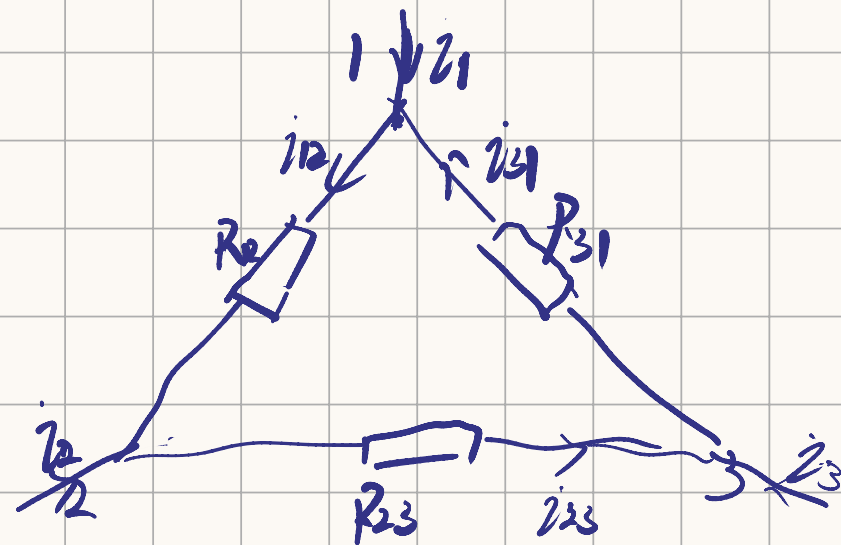


$$i_{12} = \frac{U_{12}}{R_{12}} \quad i_{23} = \frac{U_{23}}{R_{23}} \quad i_{31} = \frac{U_{31}}{R_{31}}$$

$$i_1 = i_{12} - i_{31} = \frac{U_{12}}{R_{12}} - \frac{U_{31}}{R_{31}}$$

$$i_2 = i_{23} - i_{12} = \frac{U_{23}}{R_{23}} - \frac{U_{12}}{R_{12}}$$

$$i_3 = i_{31} - i_{23} = \frac{U_{31}}{R_{31}} - \frac{U_{23}}{R_{23}}$$



$$i_1 + i_2 + i_3 = 0 \quad ①$$

$$U_{12} + U_{23} + U_{31} = 0$$

$$R_1 i_1 - R_2 i_2 = U_{12} \quad ②$$

$$R_2 i_2 - R_3 i_3 = U_{23} \quad ③$$

$$R_3 i_3 - R_1 i_1 = U_{31} \quad \rightarrow \text{why 没有}$$

因为这样会出现逻辑错误 $U_1 > U_2 > U_3$

不可以

$$\textcircled{2} \rightarrow i_2 = \frac{R_1 i_1 - U_{12}}{R_2} \quad \textcircled{5}$$

$$\textcircled{2} - \textcircled{3} \cdot (R_1 i_1 - R_2 i_2) + (R_2 i_2 - R_3 i_3) = U_{12} + U_{23}$$

$$R_1 i_1 - R_3 i_3 = U_{12} + U_{23} \quad \textcircled{6}$$

$$\textcircled{4} \rightarrow U_{12} + U_{23} = -U_{31} \quad \text{则 } R_1 i_1 - R_3 i_3 = -U_{31}$$

$$i_3 = \frac{U_{31} + R_1 i_1}{R_3} \quad \textcircled{7}$$

$$\textcircled{1} \rightarrow i_1 + \frac{R_1 i_1 - U_{12}}{R_2} + \frac{R_1 i_1 + U_{31}}{R_3} = 0$$

$$i_1 R_2 R_3 + R_3 (R_1 i_1 - U_{12}) + R_2 (R_1 i_1 + U_{31}) = 0$$

$$i_1 (R_1 R_2 + R_2 R_3 + R_1 R_3) = U_{12} R_3 - U_{31} R_2$$

$$i_1 = \frac{U_{12} R_3}{(R_1 R_2 + R_2 R_3 + R_1 R_3)} - \frac{U_{31} R_2}{(R_1 R_2 + R_2 R_3 + R_1 R_3)} \quad (2)$$

同理

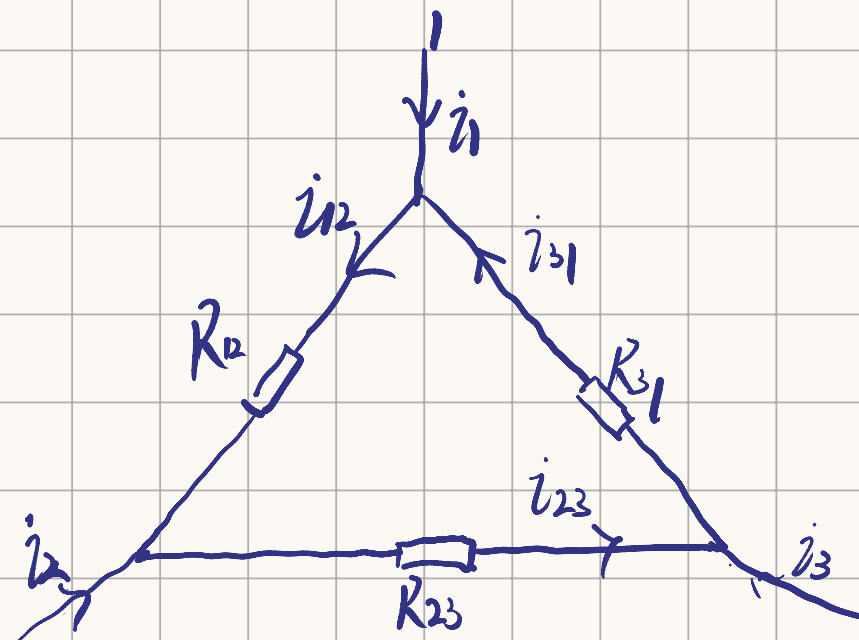
$$i_2 = \frac{U_{23} R_1}{R_1 R_2 + R_2 R_3 + R_1 R_3} - \frac{U_{12} R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3} \rightarrow a$$

$$i_3 = \frac{U_{31} R_2}{R_1 R_2 + R_2 R_3 + R_1 R_3} - \frac{U_{23} R_1}{a}$$

$$i_1 = i_{12} - i_{31} = \frac{U_{12}}{R_{12}} - \frac{U_{31}}{R_{31}}$$

$$i_2 = i_{23} - i_{12} = \frac{U_{23}}{R_{23}} - \frac{U_{12}}{R_{12}}$$

$$i_3 = i_{31} - i_{23} = \frac{U_{31}}{R_{31}} - \frac{U_{23}}{R_{23}}$$



$$i_1 = A U_{12} - B U_{31}$$

$$i_2 = C U_{23} - A U_{12}$$

$$\begin{cases} R_{12} = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3} \\ R_{31} = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2} \\ R_{23} = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1} \end{cases}$$

$$N = R_1 R_2 + R_2 R_3 + R_1 R_3$$

$$\begin{cases} R_3 = \frac{N}{R_{12}} \\ R_2 = \frac{N}{R_{31}} \\ R_1 = \frac{N}{R_{23}} \end{cases}$$

$$N = R_1 R_2 + R_2 R_3 + R_1 R_3$$

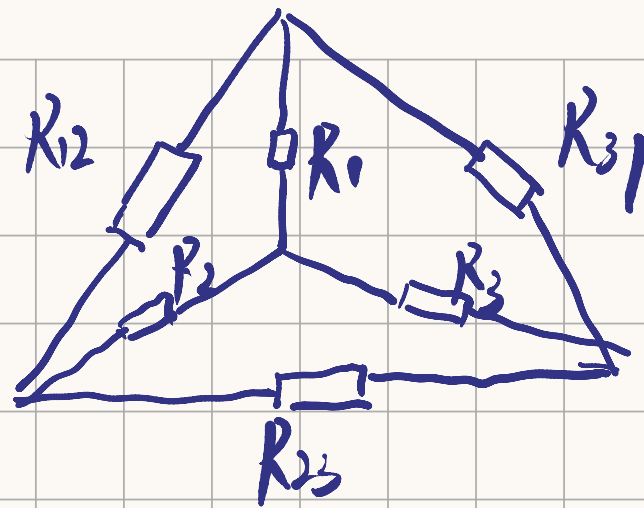
$$\begin{aligned} &= \left(\frac{N}{R_{23}}\right)\left(\frac{N}{R_{31}}\right) + \left(\frac{N}{R_{13}}\right)\left(\frac{N}{R_{12}}\right) + \left(\frac{N}{R_{23}}\right)\left(\frac{N}{R_{12}}\right) \\ &= N^2 \left(\frac{1}{R_{23} R_{31}} + \frac{1}{R_{13} R_{12}} + \frac{1}{R_{12} R_{23}} \right) \end{aligned}$$

$$N = \frac{R_{12} R_{23} R_{13}}{R_{12} + R_{23} + R_{13}}$$

$$R_3 = \frac{1}{\frac{1}{R_{12}} + \frac{1}{R_{23}} + \frac{1}{R_{31}}}$$

$$R_2 = \frac{R_{12} R_{23}}{R_{12} + R_{23} + R_{31}}$$

$$R_1 = \frac{R_{23} R_{12}}{R_{12} + R_{23} + R_{13}}$$



$\Delta \rightarrow Y$

$$R_1 = \frac{R_{12} R_{31}}{R_{12} + R_{23} + R_{31}}$$

$$R_2 = \frac{R_{12} R_{23}}{R_{12} + R_{23} + R_{31}}$$

$$R_3 = \frac{R_{31} R_{23}}{R_{12} + R_{23} + R_{31}}$$

$$R_0 = R_{12} = R_{23} = R_{31}$$

$$R_Y = R_1 = R_2 = R_3$$

$Y \rightarrow \Delta$

$$R_{12} = \frac{R R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$$R_{23} = \frac{R R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_{31} = \frac{R R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R = \frac{(R_0)^2}{3R_0} = \frac{R_0}{3} = R_2 = R_3$$

$$R_{12} = \frac{3R^2}{R} = 3R_Y = R_0$$

外3内1
图形收缩

